

Stronger Lower Bounds for Tree Covers via Cyclic Symmetry

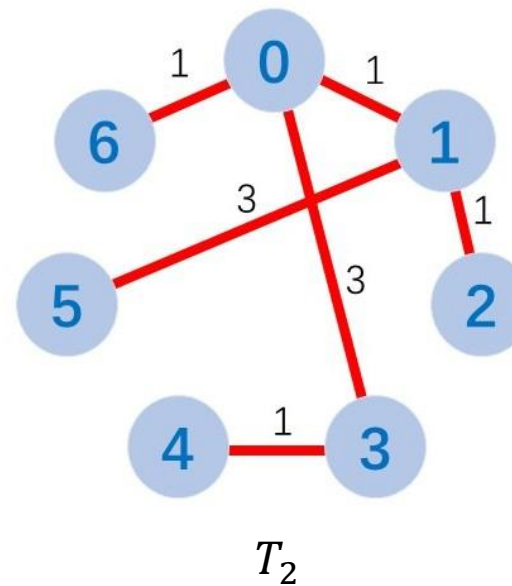
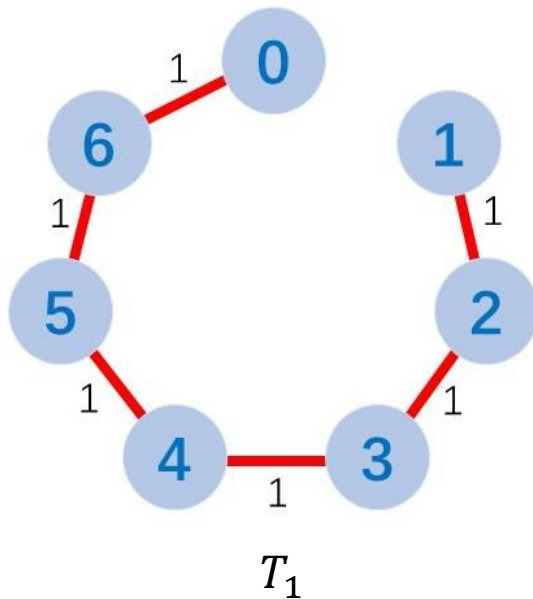
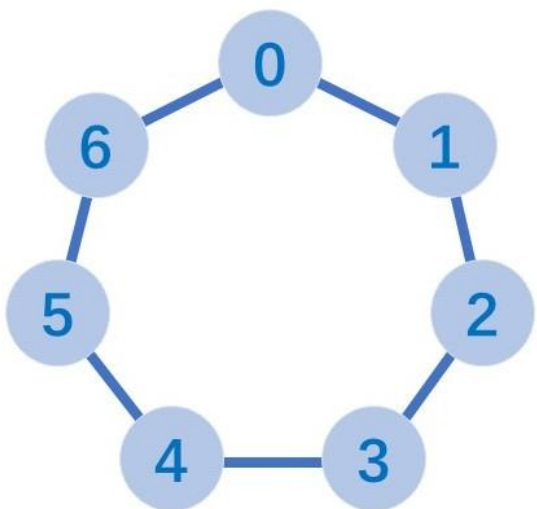
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Tree Cover

Definition (Tree Cover [Gupta, Kumar, Rastogi '01])

- **Input:** A metric space (X, d_X) on $|X| = n$ points
- **Output:** $\mathcal{T} = \{T_1, T_2, \dots, T_k\}$ of k (edge-weighted) trees on X , s.t. for any $x, y \in X$:
 - For any $i \in [k]$, $d_{T_i}(x, y) \geq d_X(x, y)$, i.e. T_i is a **dominating** tree.
 - Exist an $i \in [k]$, $d_{T_i}(x, y) \leq \alpha \cdot d_X(x, y)$. α is called the **distortion/stretch** of $\mathcal{T}$.



$\alpha = 1$

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- Tree covers provide a way to approximate general metrics by **a small collection of tree metrics**, with applications in routing [Chan, Gupta, Maggs, Zhou '16], network design [Gupta, Hajiaghayi, Räcke '06], Steiner Point Removal [Chang, Conroy, Le, Milenković, Solomon, Than '24].

The Size–Distortion Tradeoff

Reference	Size	Distortion
[RR98, Gup01]	1	$\Theta(n)$
[BFN22, Bol04]	2	$\tilde{O}(\sqrt{n}), \Omega(\log_2 n)$
[MN07, ACE+18, BFN22, Bol04]	k	$\tilde{O}(n^{1/k}), \Omega(\log_k n)$

- We mainly focus on results on general metric spaces.

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[CTX26]	k	$\Omega_k(n^{1/2^{k-1}})$

- [Chen, Tan, Xu '26] gave the first **polynomial** lower bound, by introducing a **topological** approach.

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Our Result	k	$\Omega_k(n^{1/[k(p-1)]})$

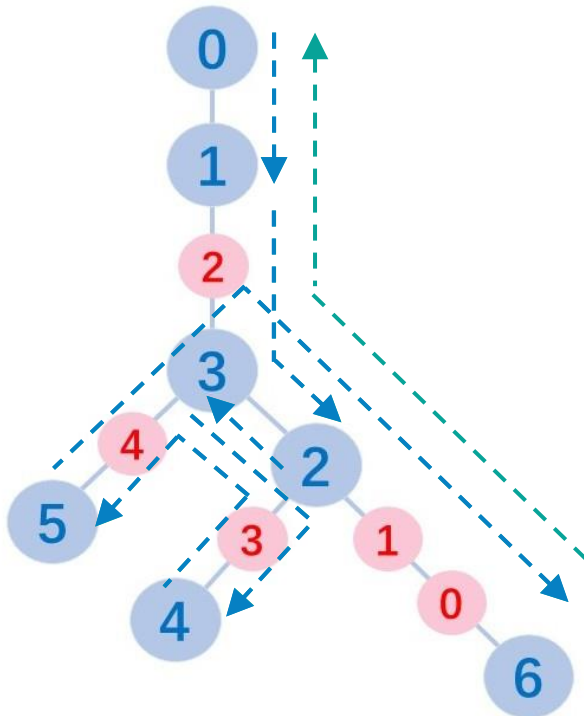
- We further develop the topological approach.
- Here p is the smallest prime s.t. $p > k$. Thus $p = O(k)$. **Gap from $\exp(k)$ to $O(k)$.**

Hard Instances for $k = 1$ and 2

Based on [Rabinovich, Raz '98], [Gupta '01] and [Chen, Tan, Xu '26]

$k = 1$: Cycle

- **Hard instance:** the unweighted **cycle** $C_n = (0, 1, \dots, n - 1)$.
- Replace each tree edge by a shortest walk on the cycle.
- Let Π_i be the induced walk from i to $i + 1$.
- Always exist a **bad** Π_i !



$$\Pi_0 = (0, 1)$$

$$\Pi_1 = (1, 2, 3, 2)$$

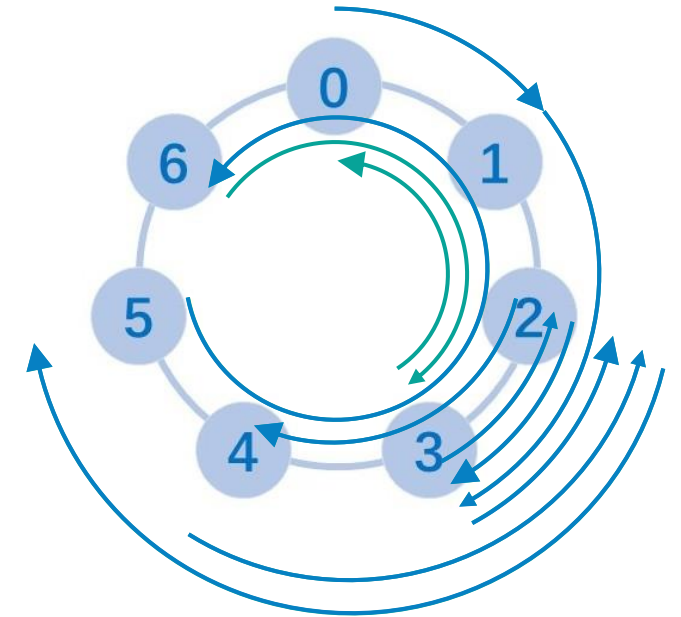
$$\Pi_2 = (2, 3)$$

$$\Pi_3 = (3, 2, 3, 4)$$

$$\Pi_4 = (4, 3, 2, 3, 4, 5)$$

$$\Pi_5 = (5, 4, 3, 2, 1, 0, 6)$$

$$\Pi_6 = (6, 0, 1, 2, 3, 2, 1, 0)$$



----- (5,6): bad edge
always exists in a cycle

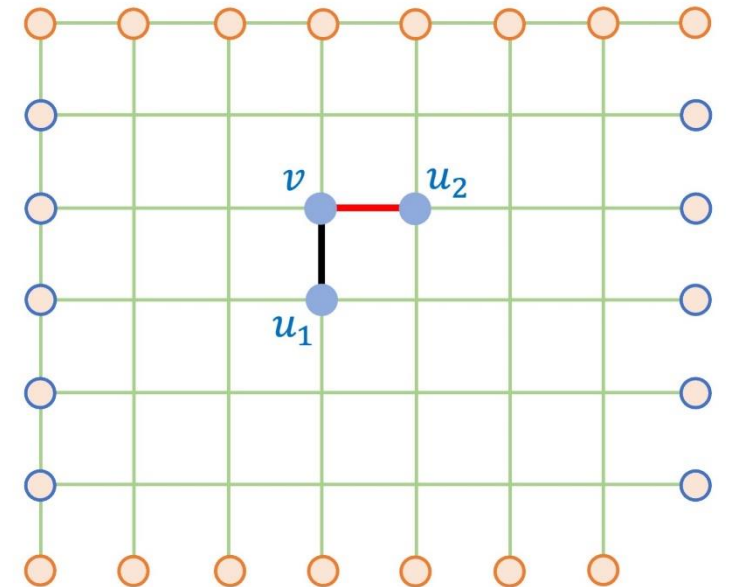
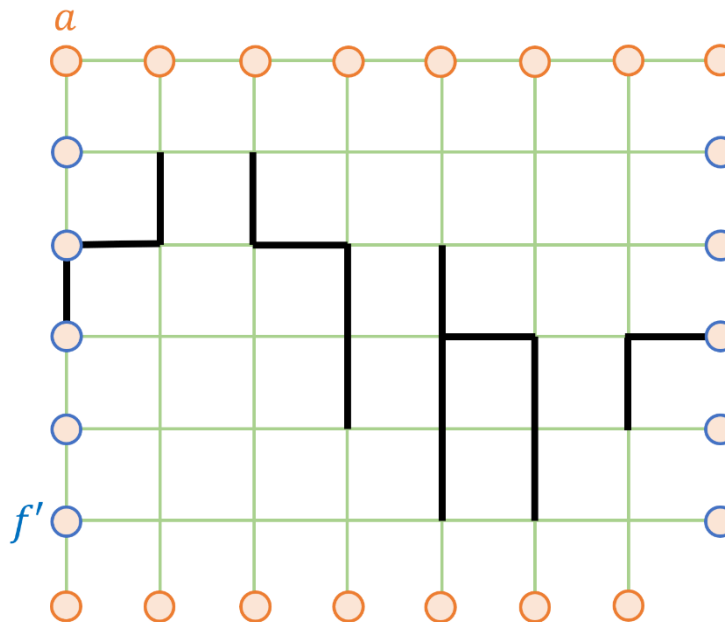
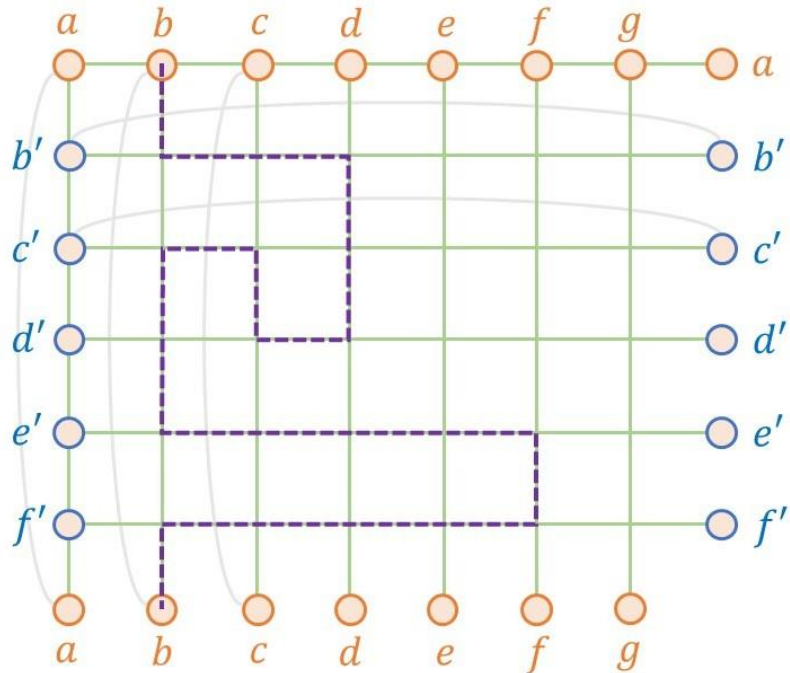
$k = 1$: Topological Intuition

- A loop going once around the cycle **cannot be contracted** to a point.
- But every continuous map from S^1 to a tree **can be contracted**, since a tree has **no topological cycle**.
- Hence a map
$$S^1 \rightarrow T \rightarrow S^1$$
- cannot preserve one full turn around the cycle.
- Such a map is called **null-homotopic** in algebraic topology.

 **Topological forcing: if the global topology cannot be carried by the approximating structure, the approximation must fail somewhere locally.**

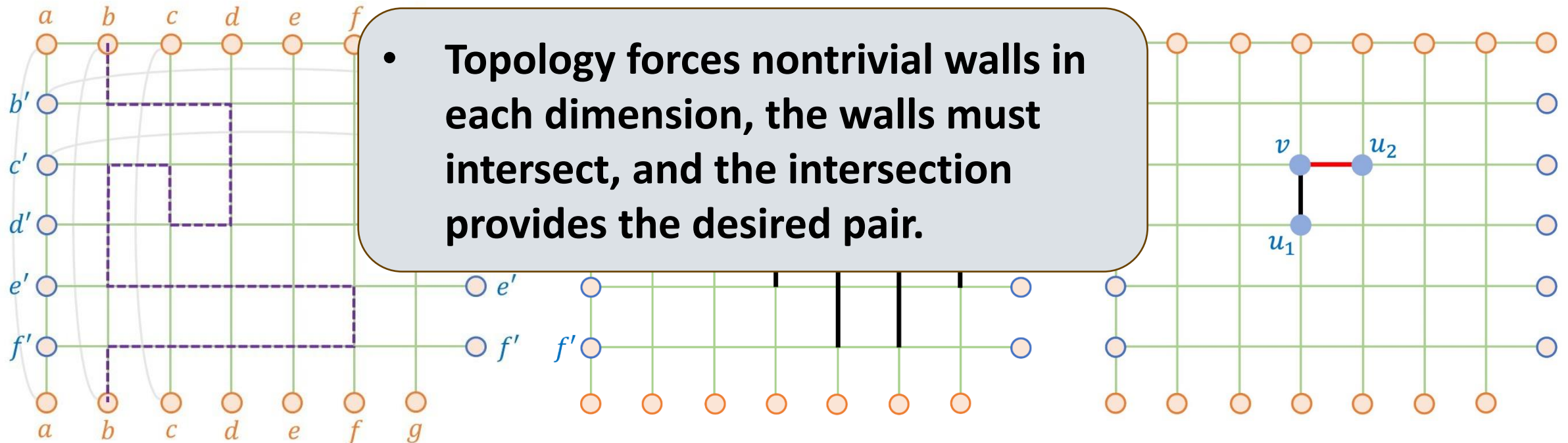
$k = 2$: Tours Grid—Plant More Cycles

- **Hard instance:** a $\sqrt{n} \times \sqrt{n}$ **tours grid**—lattices on **projective plane**.
- Every vertical/horizontal cycle contains a bad edge for T_1/T_2 .
- The union of all such edges forms a vertical/horizontal **cut**.
- The two cuts must “**intersect**” $\rightarrow$ a pair gives the lower bound $\Omega(\sqrt{n})$.



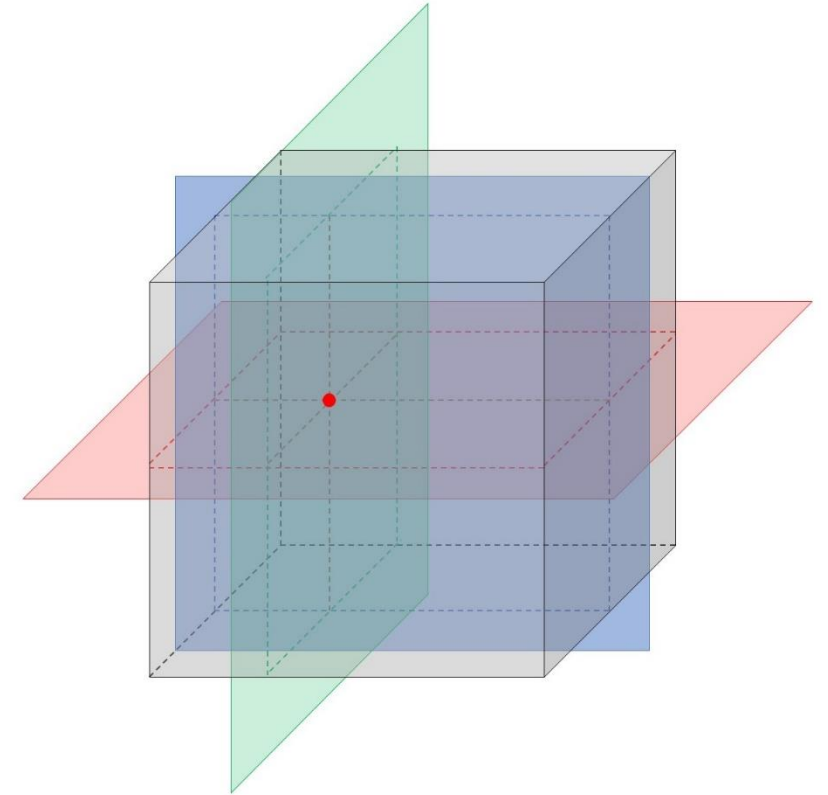
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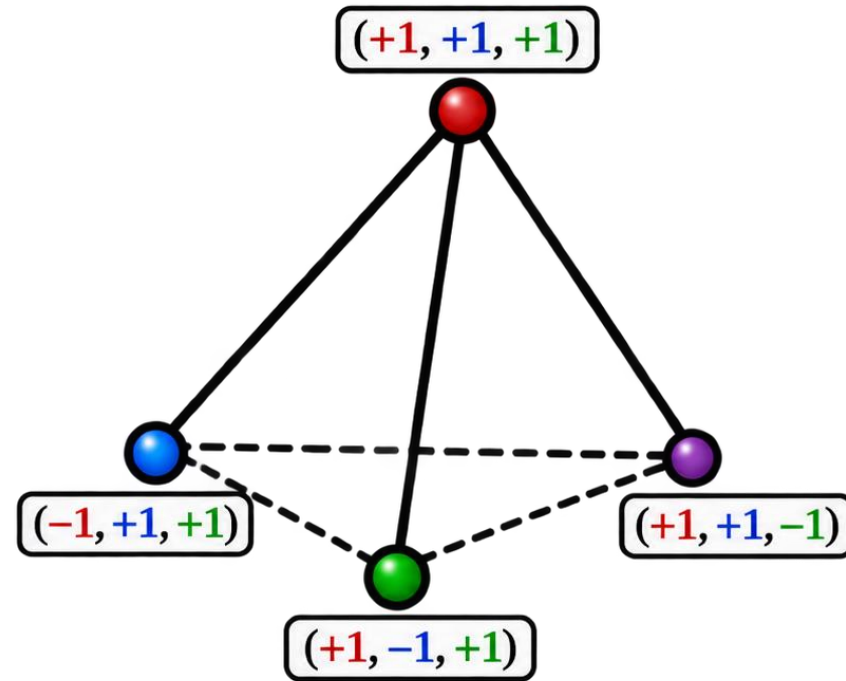
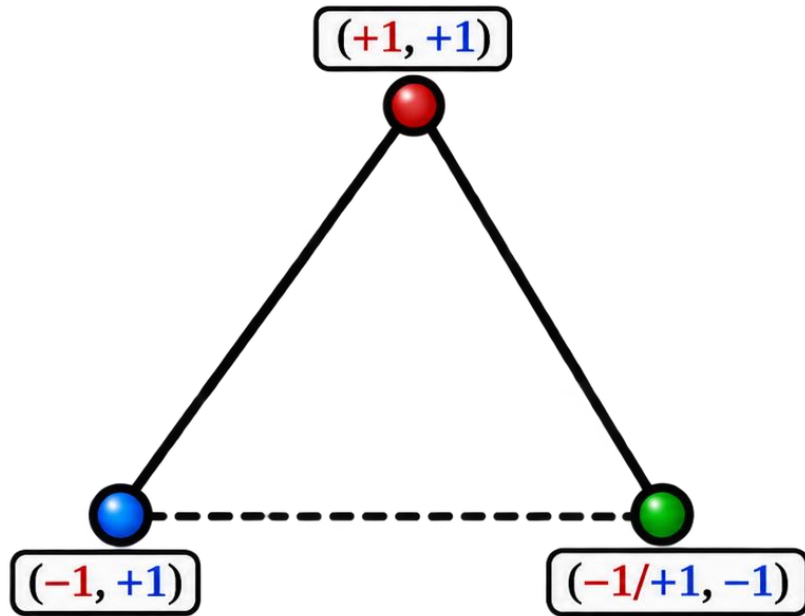
$k \geq 3$: Naive Intersection Fails

- “Hard Instance”: high-dimensional tours grid/cube
- Again, bad edges in each dimension form a cut.
- Label each vertex with $\{-1, +1\}^3$. For dimension i :
 - $+1$, if lies on the left side of the cut
 - -1 , if lies on the right side of the cut
- What we get: centrally labelled cube
- $\forall i, \exists u, v$, s.t. $u_i = -v_i$.
- What we want: complementary edge
- $\exists u, v$, s.t. $\forall i, u_i = -v_i$.



$k \geq 3$: Naive Intersection Fails

- Centrally Labeled Cube $\neq$ Complementary Edges, for $k \geq 3$.



- We still have an intersection of all dimensional cuts, but this intersection is not enough to provide the desired pair.
- Need new topological forcing!

Topological Forcing for Larger k

Based on [Chen, Tan, Xu '26]

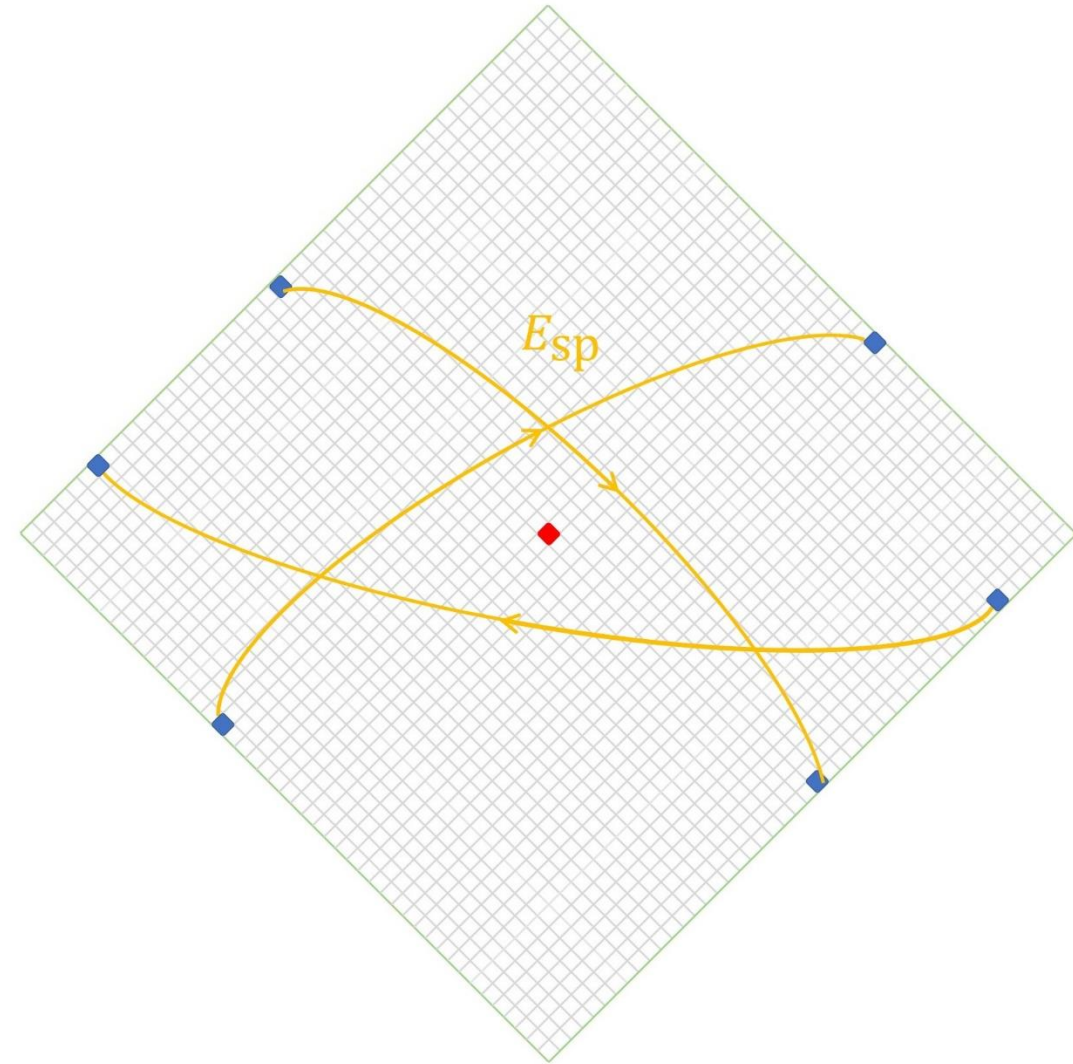
Hard Instances and Labels

Hard Instance G

- **Vertex set:** $\{x \in \mathbb{Z}^D : \|x\|_1 \leq r\}$, $r = \Theta(n^{1/D})$.
- **Ordinary edges (weight 1):** $E_0 = \{(u, v) : \|u - v\|_1 = 1\}$.
- **Special edges (weight 0):** $E_{\text{sp}} = \{(u, v) : \|u\|_1 = r, u = -v\}$.

Labels for Tree T

- $\ell_T(v)$ = the **parity** ($-1/+1$) of the special edges on **the path from the origin c to v along T** .



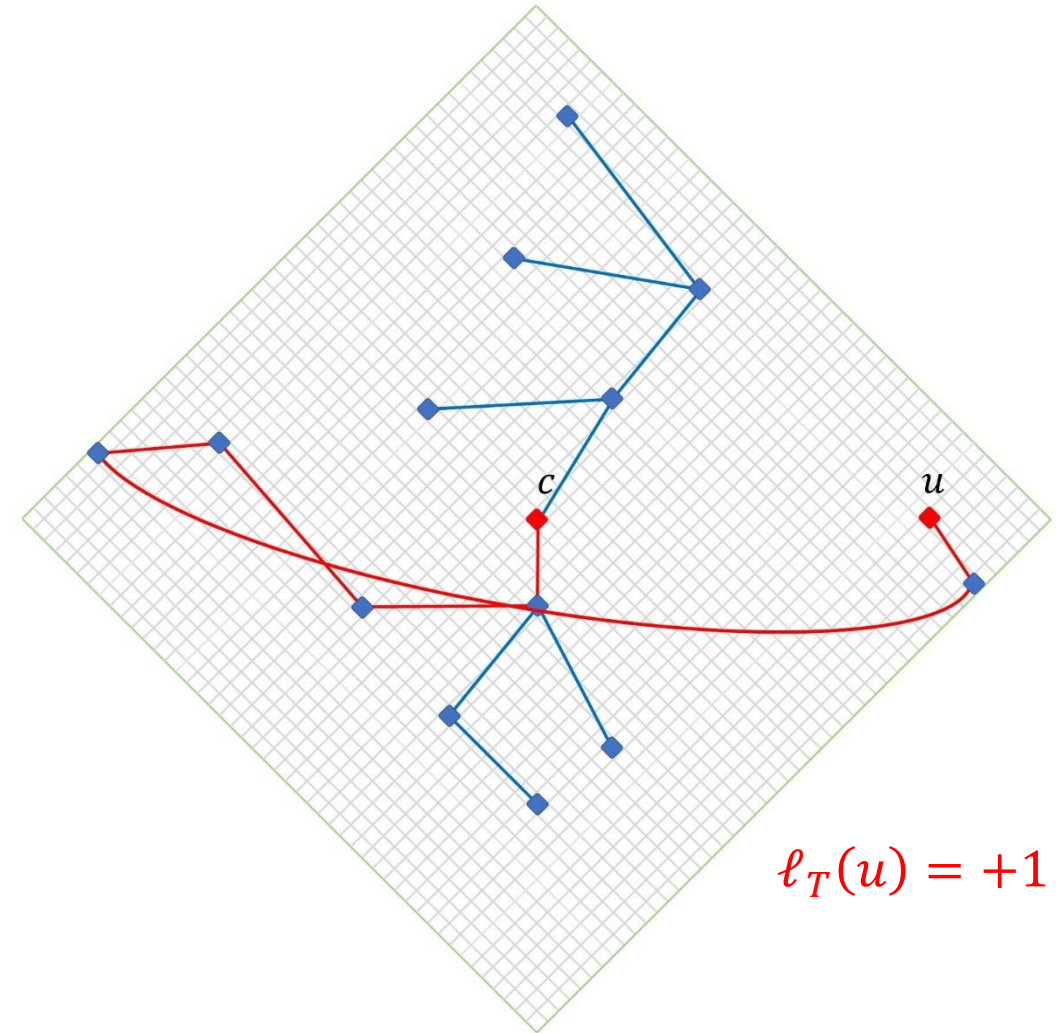
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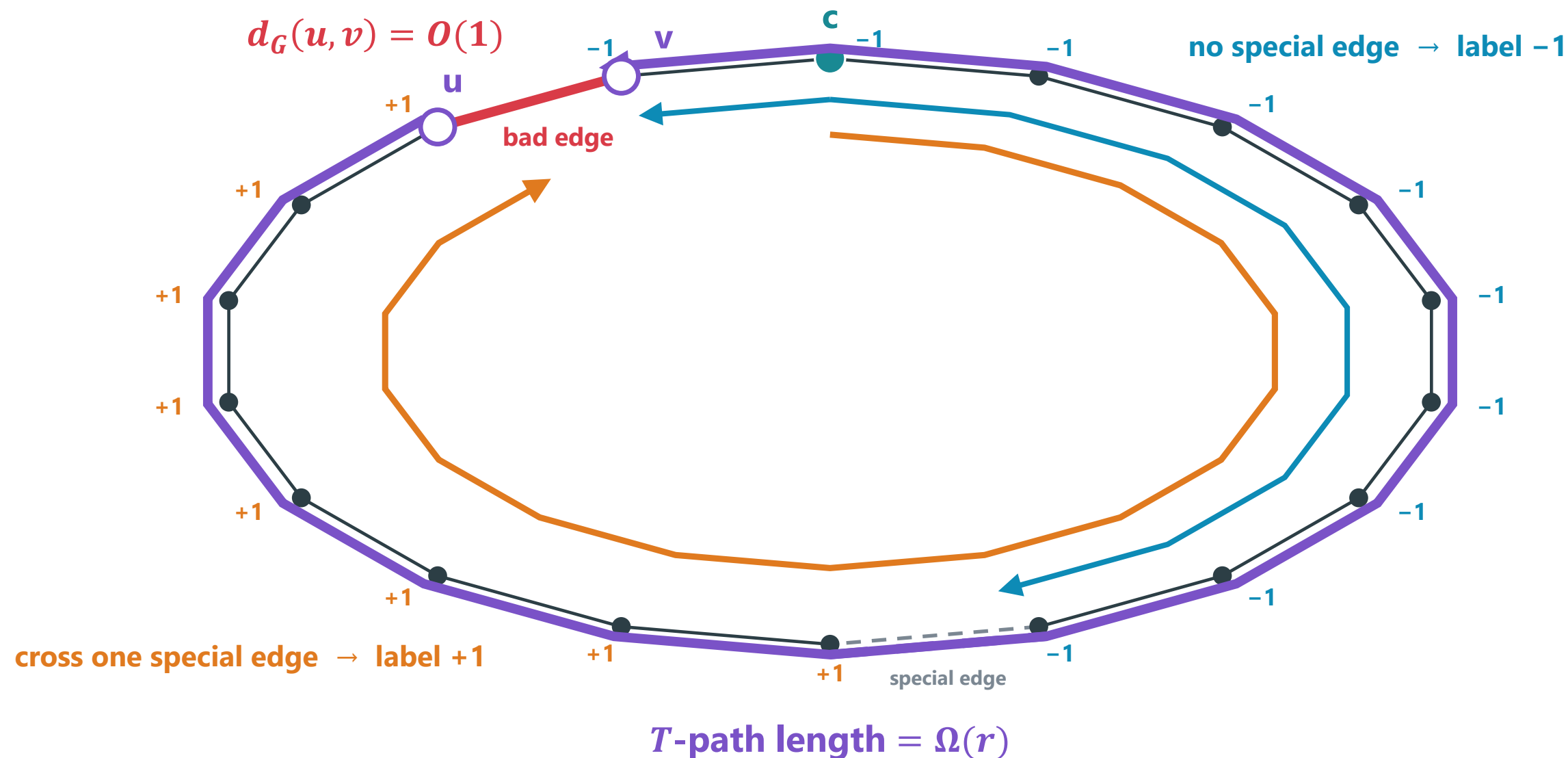
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Nearby Vertices Have Different Labels



Borsuk–Ulam Theorem and Tucker’s Lemma

- **Goal:** u, v are close in G , while have different labels in every tree (complementary pair).
- **Some Coloring Theorems – Combinatorial Fixed-Point Theorem**

Theorem (Borsuk–Ulam: Zero Formulation)

- Let $F: B^D \rightarrow \mathbb{R}^D$ be continuous and satisfy $F(-x) = -F(x)$ for $x \in \partial B^D \cong S^{D-1}$.
- Then there exists $x^* \in B^D$ such that $F(x^*) = 0$.

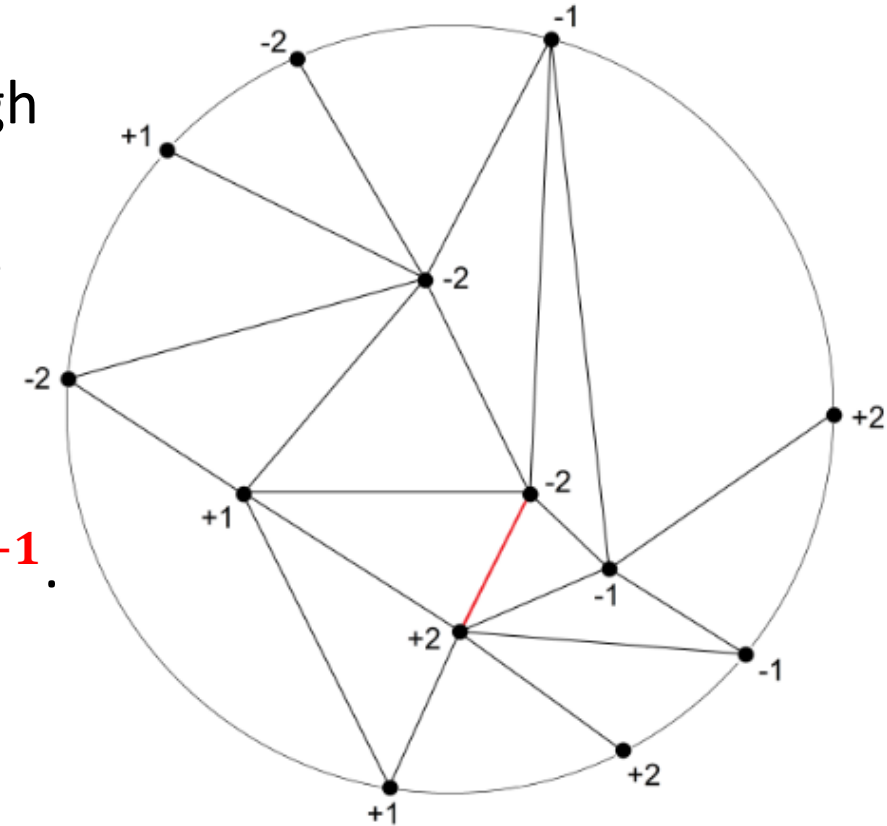
Lemma (Tucker)

- Let Γ triangulate a D -dimensional ball, with antipodally symmetric boundary triangulation.
- Suppose $\ell: V(\Gamma) \rightarrow \{\pm 1, \dots, \pm D\}$, $\ell(-v) = -\ell(v)$ for every boundary vertex v .
- Then Γ contains an edge $\{u, v\}$ such that $\ell(u) = -\ell(v)$.

Dimension Becomes the Exponent

Definition (Triangulation)

- Informally, Γ partitions a body into many triangles (in high dimension, called simplices).
 - Bounded Diameter:** same simplex $\Rightarrow$ nearby vertices
 - Discrete Vertex Labels $\Rightarrow$ Continuous Map**
 - Antipodality:** $\sigma \in \partial\Gamma \Rightarrow -\sigma \in \partial\Gamma$.
- $\{-1, +1\}^k$ contains 2^{k-1} complementary pairs, $D = 2^{k-1}$.
- Applying Tucker's Lemma:
 - u, v are in the same simplex $\Rightarrow d_G(u, v) = O(1)$.
 - $\ell(u) = -\ell(v) \Rightarrow d_{T_i}(u, v) = \Omega(r) = \Omega(n^{1/D}), \forall i \in [k]$.
 - $\alpha = \Omega_k(n^{1/2^{k-1}})$



An example of Tucker's Lemma on Wikipedia

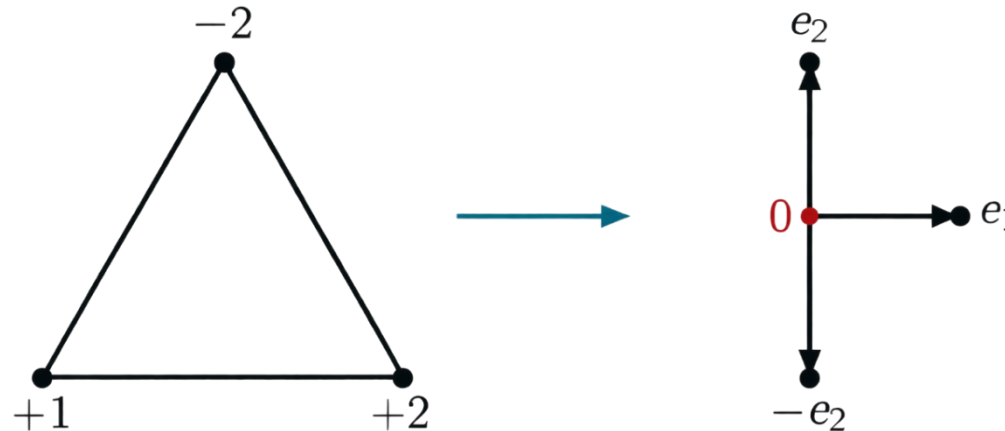
Why Is 2^{k-1} Unavoidable?

- We can see this by deriving Tucker's Lemma from Borsuk–Ulam Theorem.

- Map Tucker labels to **signed basis vectors**:

$$F(v) = \begin{cases} e_i & \ell(v) = +i \\ -e_j & \ell(v) = -j \end{cases}, \quad \forall v \in V(\Gamma).$$

- Define F on triangulation vertices and **extend it affinely on every simplex**.



- Boundary antipodality gives $F(-x) = -F(x)$ on ∂B^D .

Why Is 2^{k-1} Unavoidable?

- Borsuk–Ulam gives $x^* \in \sigma$ with $F(x^*) = 0$. Writing

$$x^* = \sum_{v \in V(\sigma)} \lambda_v v, \quad \lambda_v \geq 0, \quad \sum_{v \in V(\sigma)} \lambda_v = 1.$$

- We get

$$0 = F(x^*) = \sum_{v \in V(\sigma)} \lambda_v F(v).$$

- If σ contained **no complementary pair**, then for each coordinate i **at most one** of $e_i, -e_i$ could appear. Such vectors can not have 0 in their convex hull.
- Borsuk–Ulam forces a zero; the encoding turns the zero into a complementary pair.
- **We need one dimension for every complementary pair! Thus 2^{k-1} is unavoidable.**

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- Can we do more cleverly? Apply the topological forcing while using fewer dimension.

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From Antipodal Symmetry to Cyclic Symmetry

Our Approach [Huang '26]

Binary Labels vs p -ary Labels

Binary labels: a prescribed partner

- For $x \in \{-1, +1\}^k$, the **only** label differing from x in every coordinate is $-x$.
- Thus the good-pair relation is a perfect matching with 2^{k-1} antipodal pair types.

- A direct Tucker encoding has to distinguish these 2^{k-1} pair types.

- **Relaxation:** any all-coordinate-different partner is enough, so lower-dimensional forcing may be possible.

p -ary labels: many compatible partners

- For $x \in \mathbb{Z}_p^k$, each coordinate of y has $p - 1$ valid choices.
- Thus x has $(p - 1)^k$ **partners** satisfying $y_j \neq x_j, \forall j \in [k]$.

- If changing coordinate j already means being far in tree T_j , then **any** such compatible partner works.

A Balanced View of Borsuk–Ulam

- With binary joint labels $\ell(v) \in \{-1, +1\}^k$, a Borsuk–Ulam zero inside some simplex σ gives barycentric weights $\{\lambda_v\}$ such that

$$\sum_{v \in V(\sigma)} \lambda_v \ell(v) = 0.$$

- Therefore every coordinate is **exactly balanced**:

$$\sum_{\ell_j(v)=+1} \lambda_v = \sum_{\ell_j(v)=-1} \lambda_v = \frac{1}{2}, \quad \forall j \in [k].$$

- Sample U, V **independently** from $V(\sigma)$ according to $\{\lambda_v\}$. Then

$$\mathbb{E}[\#\{j: \ell_j(U) \neq \ell_j(V)\}] = \sum_{j=1}^k \Pr[\ell_j(U) \neq \ell_j(V)] = \frac{1}{2} \times k = \frac{k}{2}.$$

- There exists a pair of vertices differing in at least half of coordinates. Not enough!**

A Balanced View of Borsuk–Ulam

- If we use $\mathbb{Z}_p$ labels. **Suppose** that inside one small simplex σ , topology could force

$$\sum_{\ell_j(v)=a} \lambda_v = \frac{1}{p}, \quad \forall j \in [k], a \in \mathbb{Z}_p.$$

- Sample U, V independently according to the same weights. For every tree T_j ,

$$\Pr[\ell_j(U) = \ell_j(V)] = \sum_{a \in \mathbb{Z}_p} \left(\frac{1}{p}\right)^2 = \frac{1}{p}.$$

- Therefore

$$\Pr[\exists j \in [k]: \ell_j(U) = \ell_j(V)] \leq \frac{k}{p}.$$

- Choose $p > k$. Then with **positive probability**, $\forall j \in [k], \ell_j(U) \neq \ell_j(V)$.

- **Hence there exists a pair differing in every coordinate.**

Dold's Theorem: $\mathbb{Z}_p$ -Equivariant Borsuk–Ulam Theorem

Theorem (Dold: $\mathbb{Z}_p$ -Equivariant Zero Principle)

- Let g generate an action of $\mathbb{Z}_p$ on a D -dimensional ball B^D and on $\mathbb{R}^D$. Assume that the action is **free** on ∂B^D and on $\mathbb{R}^D \setminus \{0\}$, while **0** is **fixed**.
- Suppose a **continuous** map $F: B^D \rightarrow \mathbb{R}^D$ is **$\mathbb{Z}_p$ -equivariant** on the boundary, i.e.
$$F(gx) = gF(x) \quad (x \in \partial B^D).$$
- Then there exists $x^* \in B^D$ such that $F(x^*) = 0$.

Borsuk–Ulam ($\mathbb{Z}_2$ -Equivariant)

$$x \mapsto -x$$

- antipodal symmetry.

Here

$$x \mapsto gx, \quad g^p = \text{id}$$

- cyclic symmetry.

Encode One $\mathbb{Z}_p$ Label by a Centered Simplex

- We want a geometric encoding in which **zero means uniform balance**.
- For $a \in \mathbb{Z}_p$, define the vertices of a centered $(p-1)$ -simplex by

$$q_0 = -\frac{1}{p}\mathbf{1}, \quad q_a = e_a - \frac{1}{p}\mathbf{1} \quad (a \in \mathbb{Z}_p \setminus \{0\}).$$

- Then $q_0 + q_1 + \cdots + q_{p-1} = \mathbf{0}$, and the cyclic action permutes the vertices $gq_a = q_{a+1}$.
- Hence if $w_a \geq 0$, $\sum_a w_a = 1$, and

$$\sum_{a \in \mathbb{Z}_p} w_a q_a = \mathbf{0},$$

- then necessarily

$$w_0 = w_1 = \cdots = w_{p-1} = \frac{1}{p}.$$

From p -ary Labels to an Equivariant Map

- For k trees, let every triangulation vertex carry a joint label

$$\ell(v) = (\ell_1(v), \dots, \ell_k(v)) \in \mathbb{Z}_p^k.$$

- Use one centered simplex for each tree and define

$$F(v) = (q_{\ell_1(v)}, \dots, q_{\ell_k(v)}) \in (\mathbb{R}^{p-1})^k = \mathbb{R}^D, \quad D = k(p-1).$$

- Extend F affinely over every simplex of the triangulation.

- If the labels satisfy

$$\ell_j(gv) \equiv \ell_j(v) + 1 \pmod{p}, \quad \forall j \in [k], v \in \partial B^D,$$

- then $gq_a = q_{a+1}$ gives

$$F(gx) = gF(x) \quad (x \in \partial B^D).$$

- Thus the $\mathbb{Z}_p$ -equivariant zero principle applies and **forces** some x^* with $F(x^*) = \mathbf{0}$.

New Topological Forcing: Balanced Simplex

- Let x^* be a zero of F , contained in a simplex σ , and write $x^* = \sum_{v \in V(\sigma)} \lambda_v v$. Since F is affine on σ ,

$$0 = F(x^*) = \sum_{v \in V(\sigma)} \lambda_v F(v).$$

- Hence, for every tree T_j and every $a \in \mathbb{Z}_p$,

$$\sum_{\ell_j(v)=a} \lambda_v = \frac{1}{p}.$$

Lemma (Balanced Simplex)

- Under the $\mathbb{Z}_p$ boundary condition, some simplex σ is uniformly balanced in every tree label. Since $p > k$, the p -ary sampling argument then gives two vertices $u, v \in V(\sigma)$ such that

$$\ell_j(u) \neq \ell_j(v), \quad \forall j \in [k].$$

Building a Cyclic Geometry

- The Balanced Simplex Lemma requires a **boundary with a cyclic $\mathbb{Z}_p$ -symmetry**.
- Represent one block by a **zero-sum p -tuple**:

$$\hat{x}_0 = - \sum_{a=1}^{p-1} x_a, \quad \hat{x}_a = x_a \quad (1 \leq a \leq p-1).$$

- Let g cyclically shift the completed coordinates:

$$(\hat{x}_0, \dots, \hat{x}_{p-1}) \mapsto (\hat{x}_{p-1}, \hat{x}_0, \dots, \hat{x}_{p-2}).$$

- Thus $g^p = \text{id}$.

- For k trees:

$$\mathbb{R}^D = (\mathbb{R}^{p-1})^k, \quad D = k(p-1).$$

Hard Instance: Cyclic Boundary Quotient

- Define

$$\rho(x) = \frac{1}{2} \sum_{j=1}^k \sum_{a \in \mathbb{Z}_p} |\hat{x}_{j,a}|, \quad B_r = \{x: \rho(x) \leq r\}, \quad \Lambda_r = B_r \cap \mathbb{Z}^D.$$

Hard Instance G

- Vertex set: Λ_r , $r = \Theta(n^{1/D})$.
- Ordinary edges (**weight 1**): $E_0 = \{(u, v): \rho(u - v) = 1\}$.
- Special edges (**weight 0**): $E_{\text{sp}} = \{(u, gu): \rho(u) = r\}$.
- One special transition = one cyclic phase shift.

Each Tree Induces a Cyclic Winding Label

- Fix a dominating tree T rooted at the origin c . For a special edge, $x \rightarrow gx$: forward;
 $gx \rightarrow x$: backward.
- For a path P in G , define its **winding number**
$$\text{wind}(P) = \#(\text{forward}) - \#(\text{backward}) \pmod{p}.$$
- This gives a phase label
$$\ell_T(x) = \text{wind}(P^T(c, x)) \in \mathbb{Z}_p.$$
- The key identity is
$$\ell_T(gx) \equiv \ell_T(x) + 1 \pmod{p}, \quad \forall x \in \partial B_r.$$
- This is exactly the equivariance required by the Balanced Simplex Lemma.

Apply the Balanced Simplex Lemma

- Choose a triangulation Γ of B_r such that: (1) $V(\Gamma) \subseteq \Lambda_r$; (2) **boundary g -invariant**: $\sigma \in \partial\Gamma \Rightarrow g\sigma \in \partial\Gamma$; (3) **bounded diameter**: $u, v \in \sigma \Rightarrow \rho(u - v) \leq C_\Gamma = O_k(1)$.
- The hypotheses are now satisfied
$$D = k(p - 1), \quad \ell_j(gx) = \ell_j(x) + 1, \forall j \in [k], x \in \partial B_r,$$
- and the boundary action is **g -equivariant**.
- **Applying the Balanced Simplex Lemma**, there exist $\sigma \in \Gamma$ and $u, v \in V(\sigma)$ such that
$$\ell_j(u) \neq \ell_j(v), \forall j \in [k].$$
- Moreover, $d_G(u, v) = O_k(1)$.
- **The topology is now finished.**

Orbit Points Are Far Apart

Lemma (Cyclic Displacement)

- For every $x \in \mathbb{R}^D$ and every $s \in \{1, \dots, p-1\}$,

$$\rho(x - g^s x) \geq \frac{2}{p} \rho(x).$$

- Hence, for $x \in \partial B_r$, $\rho(x - g^s x) \geq 2r/p$.

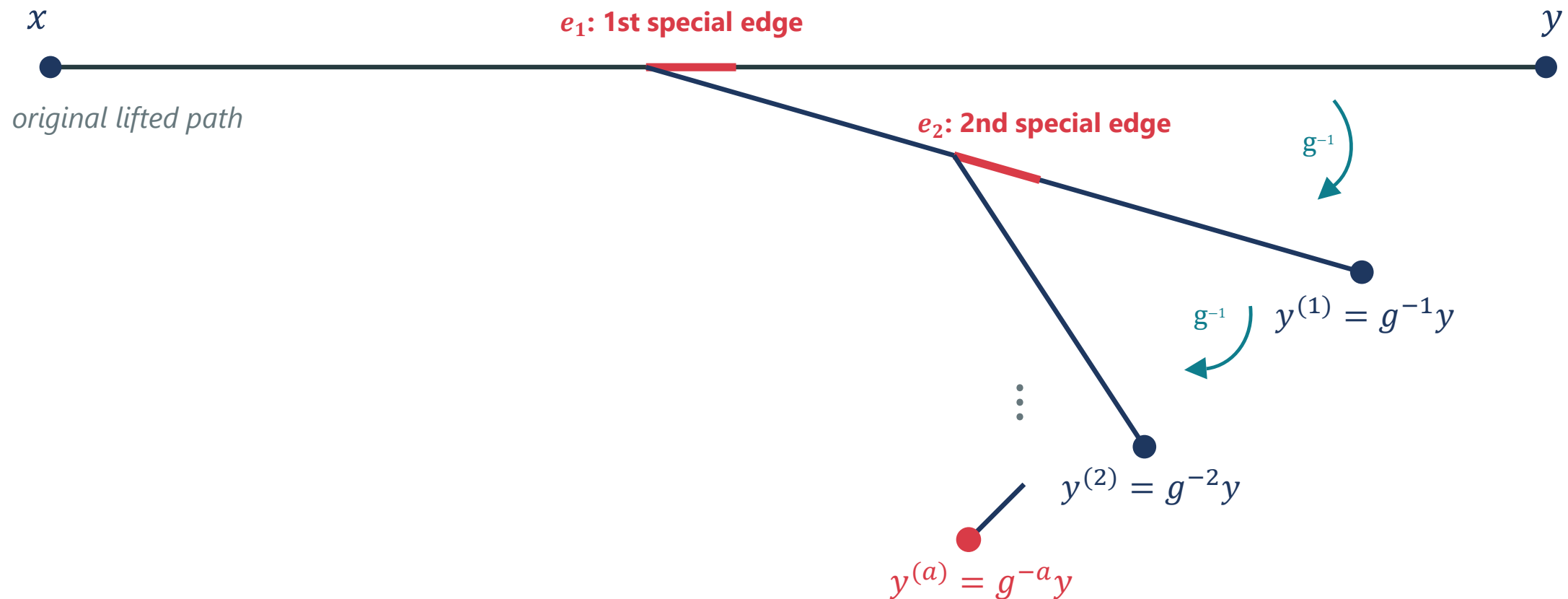
Proof Sketch

- For fixed $s \neq 0$, $a \mapsto a - s$ is a single p -cycle. Hence, for each block, the p cyclic shifts of the completed coordinates have total variation at least 2 times the block mass. Averaging over the p shifts gives $\rho(x - g^s x) \geq 2\rho(x)/p$ for every $s \not\equiv 0 \pmod{p}$.
- This is the cyclic analogue of the binary fact that x and $-x$ are far apart.
- Key intuition:** different labels should correspond to large geometric separation.

Cyclic Unwinding

Lemma (Cyclic Unwinding)

- If a path P goes from x to y and has winding $a \in \mathbb{Z}_p$, then there is an **ordinary-edge-only** path from x to $y^{(a)} = g^{-a}y$ with **exactly the same weighted length**.



Different Phase $\Rightarrow$ Large Tree Distance

- Let $a = \ell_T(v) - \ell_T(u) \neq 0$. By **unwinding**,
$$d_T(u, v) \geq \rho(u - g^{-a}v).$$

Case1

- Assume $\rho(v) \geq r/2$. Since u, v lie in one simplex, $\rho(u - v) \leq C_\Gamma$. Hence
$$d_T(u, v) \geq \rho(v - g^{-a}v) - \rho(u - v) \geq 2\rho(v)/p - C_\Gamma \geq r/p - C_\Gamma.$$

Case2

- Suppose $\rho(u), \rho(v) < r/2$. Non zero winding forces the path to touch the boundary, so it must go out and return:

$$d_T(u, v) \geq (r - \rho(u)) + (r - \rho(v)) > r.$$

- For $r \geq 2pC_\Gamma$,

$$d_T(u, v) \geq \frac{r}{2p}, \quad \forall j \in [k].$$

Putting Everything Together

- The **Balanced Simplex Lemma** gives u, v in the same simplex with different labels in every tree.
- Triangulation has **bounded diameter**, giving $d_G(u, v) \leq C_\Gamma$.
- For every $j \in [k]$, $d_{T_j}(u, v) \geq r/(2p)$.

$$\alpha \geq \frac{r}{2pC_\Gamma} = \Omega_k(r)$$

$$D = k(p - 1), \quad |V(G)| = \Theta_k(r^D)$$

Choose $r = \Omega_k(n^{1/D})$ gives $\alpha = \Omega_k(n^{1/[k(p-1)]})$

Summary

	Chen–Tan–Xu	Ours
Per-Tree Label	$\mathbb{Z}_2$	$\mathbb{Z}_p$
Desired Partner	Unique Antipode	Many Compatible Partners
Topological Output	Complementary Pair	Balanced Simplex
Extraction	Direct	Sampling + Union Bound
Dimension	2^{k-1}	$k(p-1) = O(k^2)$

- The key change is not the alphabet itself, but relaxing from finding a prescribed partner to exploiting p -ary compatibility through balance.

Open Problems

- **For general metric spaces:** upper bound = $\tilde{O}(n^{1/k})$, lower bound = $\Omega_k(n^{1/O(k^2)})$.
- For $k \geq 3$, obtaining tight (or nearly tight) bounds remains open.
- Many questions remain open for **special metric families**, such as
 - Euclidean and doubling metrics [Chan '98], [Bikeev, Kupavskii, Turevskii '26], [Le, Milenković, Solomon, Zhang '26] and [Chen, Tan, Xu '26];
 - planar and minor-free graph metrics [Le, Pham, Than, Tran '26].
- How much can additional geometric or graph structure improve the size–distortion tradeoff?
- We expect the stretch reduces to $O(1)$ at some point as the number of trees increasing.

Thank you for listening!

- Thanks **Hangyu Xu** for generously sharing his slides of their work [\[Chen, Tan, Xu '26\]](#) as material for making these slides.
- Pictures in Page 2, 8, 10, 11 and 14 are taken from [Hangyu's slides](#).